Sumesh Surendran Letha

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Open to relocate | Exton, PA

PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with a decade of experience in designing and delivering enterprise-grade distributed systems and cloud- native microservices in healthcare and aviation domains. Deep expertise in Java, Spring Boot, Python, React, Apache Kafka and AWS with a strong expertise of architecture HIPAA-complaint, high throughput data pipelines at Siemens Healthineers. Currently completing M.S in Computer Science at Villanova University with active research in LLM-powered automation.

EDUCATION

Villanova University | *College of Liberal Arts and Sciences* — Villanova, PA

Master of Science in Computer Science — **May 2026**

Relevant Coursework: *Advanced Algorithms* | *Data Structures* | *Artificial Intelligence* |*Operating Systems*

Cochin University of Science and Technology | *Bachelor of Technology in Computer Science and Engineering* — Kochi, India

ACADEMIC & RESEARCH EXPERIENCE

Villanova University — *Villanova, PA*

Research Assistant — **Aug 2024 – Present**

- Architected scalable AI-driven software framework with voice-to-action automation framework integrating API gateway, pre/post-processing, and verification microservices, thus reducing cross-app navigation steps.
- Engineered a full-stack AI-powered clinical training platform for nursing students, enabling real-time scenario-based conversations with multi-role AI bots via React frontend, Spring Boot/Node.js backend, OpenAI API integration, PostgreSQL, and Socket.IO for live session observation.
- Designed and prototyped BridgeSafe, an event-driven IoT system for real-time overheight vehicle detection using Raspberry Pi 5, HC-SR04 ultrasonic sensing, and an MQTT/WebSocket pipeline on AWS EC2, achieving 287ms median end-to-end latency and 100% message delivery reliability across 100+ concurrent detector nodes.
- Authored and submitted an IEEE AIIoT 2026 conference paper on a four-layer edge-to-cloud IoT architecture integrating Eclipse Mosquitto, Spring Boot, and WebSocket push notifications for municipal bridge safety monitoring, demonstrating a cost-effective solution.

PROFESSIONAL EXPERIENCE

Siemens Medical Solutions Inc — *Newark, DE*

Senior Software Engineer — **Oct 2019 – Sep 2023**

- Architected distributed Java Spring Boot microservices adhering to SOLID design principles and Gang-of-Four patterns enabling modular, maintainable services across 200+ hospital systems.
- Led design of Acumulus & Auto QC systems, demonstrating expertise in large-scale system architecture and fault-tolerant distributed systems design with 99.9% uptime SLAs for clinical laboratory operations.
- Designed and deployed cloud-native data integration solutions leveraging AWS ECS, EMR, S3, and event-driven architecture to establish reliable, high-throughput data workflows processing 5M+ medical records/day.
- Developed Acumulus and Aptio Notifications platforms compliant with HIPAA PHI/PII standards, leveraging Apache Kafka for fault-tolerant, ordered event delivery and AWS ElastiCache for Redis, achieving <50ms read latency at peak load.
- Automated CI/CD pipelines with integrated networking configurations using VPC subnets, security groups, load balancers, and API Gateway routing, reducing deployment cycle from 2 weeks to 2 days.
- Engineered scalable backend services with Kubernetes and Terraform Infrastructure-as-Code, across dev, staging, and production environments, supporting zero-downtime deployments and auto-scaling under variable clinical workloads.

Siemens Healthcare Pvt Ltd — *Bangalore, India*

Senior Software Engineer — **May 2017 – Oct 2019**

- Optimized large-scale medical data platform performance by 30%, systematically resolving distributed system bottlenecks across data ingestion, processing, and query layers serving 50+ enterprise clients.
- Architected and implemented a comprehensive data lake management system (syngo Lab Data Manager) that consolidated patient sample and result records, enabling scalable data governance and accessibility across enterprise systems.
- Designed robust data models and implemented fault-tolerant mechanisms to ensure 99.95% data durability and consistency, under peak workloads for real-time clinical reporting and analytics at scale.

IBS Software Pvt Ltd *Thiruvananthapuram, India*

Clients*: Virgin Atlantic, Emirates, Jet Airways, Air France/KLM*

Technical Lead **May 2011 – May 2017**

- Directed development of the Leave Management system, implementing slot management, entitlement creation, leave assignment, carry-over logic, and auto-allocation for Flight and Cabin Crew which was built using Core Java, J2EE, Spring, Hibernate, and XML parsing.
- Built the Subsistence Payment module automating meal payment generation for flight crews based on layover station and hotel rates, producing structured XML payouts directly to crew cards which was developed using Core Java, J2EE, Struts, web services, and XML generation
- Developed Feeds and IOB modules for a Cosmic Radiation Monitoring system within the iFlight Crew aviation platform, processing multi-variable flight data (sector, altitude, ascent/descent rates) to track and report crew radiation exposure against rolling 12-month regulatory compliance thresholds.
- Awarded the Customer Champion title and recognized twice as Team Champion for exceptional problem-solving abilities and outstanding teamwork.

SKILLS

Languages: Java, Python, HTML/CSS, Bootstrap, JavaScript, SQL, NoSQL
Frontend: Angular, TypeScript, React, Node.js
Frameworks: Spring Boot, Spring MVC, Hibernate/JPA, Event-driven architecture, Apache Kafka
Cloud & DevOps: AWS (EC2, ECS, Fargate, EKS, S3, EMR, Elasticache for Redis Redshift, API Gateway, IAM, VPC), Kubernetes, Docker, Terraform, Jenkins, Azure DevOps, GitHub Actions
Database: Oracle, MySQL, SQL Server, SQLite, DynamoDB, Mongo DB, Data Lake
Architecture: Microservices, RESTful APIs, Event-Driven Architecture, Distributed Systems, SOLID Design Principles, Design Patterns (Factory, Observer, Strategy, Singleton, Repository)
Networking: AWS VPC, Subnets, NAT Gateway, Security Groups, Load Balancers, API Gateway, DNS
AI & ML Tools: OpenAI GPT-4, Google Gemini, LLM integration, Cursor, GitHub Copilot, Claude Code
Domain: HIPAA, PHI/PII compliance, DICOM, HL7, RxNorm, NPI | Agile, Scrum, CI/CD

CERTIFICATIONS

AWS Certified Solutions Architect – Associate| SAA-C03
AWS Certified Cloud Practitioner| CLF-C02

AWARDS

IBS Software **Customer Champion**
Two Time IBS Software **Team Champion**